Probability - Homework 3

B13902022 Yu-Chi,Lai

2.5-5

Recall that:

- $M(0) = E[e^{0X}] = E[1] = 1$
- $M'(0) = \mu = E[X]$
- $M''(0) = E[X^2]$

Thus, we obtain:

$$R'(0) = \frac{M'(0)}{M(0)} = \frac{\mu}{1} = \mu$$

$$R''(0) = \frac{M(0)M''(0) - [M'(0)]^2}{[M(0)]^2} = \frac{1 \cdot E[X^2] - (E[X])^2}{1} = E[X^2] - (E[X])^2 = \sigma^2$$

completes the proof

2.5-6

(a)

Suppose the probability of success be p, X be the r.v. standing for the number of success.

$$M(t) = P(X=0) \cdot e^{0t} + P(X=1) \cdot e^{1t} = (1-p) + p \cdot e^t$$

Then we have:

$$R(t) = \ln(1 - p + pe^t)$$

$$R'(t) = \frac{pe^t}{1 - p + pe^t}$$

$$R''(t) = \frac{(1 - p + pe^t)pe^t - (pe^t)^2}{(1 - p + pe^t)^2}$$

By the results in 2.5-5, $\mu = R'(0) = p$, $\sigma^2 = R''(0) = p(1-p)$

(b)

Suppose the probability of success be p, X be the r.v. standing for the number of success, and n be the count of the trials.

$$
\begin{aligned}
M(t) &= \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} e^{kt} \\
&= \sum_{k=0}^{n} \binom{n}{k} (pe^t)^k (1-p)^{n-k} \\
&= (1-p+pe^t)^n
\end{aligned}
$$

The last equation comes from binomial theorem. And the first and second derivatives are:

$$
\begin{aligned}
M'(t) &= n(1-p+pe^t)^{n-1} pe^t \\
M''(t) &= n(n-1)(1-p+pe^t)^{n-2}(pe^t)^2 + n(1-p+pe^t)^{n-1} pe^t
\end{aligned}
$$

Hence, since $R(t) = \ln M(t) = n \ln(1-p+pe^t)$,

$$
\begin{aligned}
R'(t) &= \frac{npe^t}{1-p+pe^t} \\
R''(t) &= \frac{npe^t(1-p)}{(1-p+pe^t)^2}
\end{aligned}
$$

By the results in 2.5-5, $\mu = R'(0) = np$ and $\sigma^2 = R''(0) = np(1-p)$.

(c)

Suppose the probability of success be p, X be the r.v. standing for the number of trials to observe one success.

$$
\begin{aligned}
M(t) &= \sum_{k=1}^{\infty} e^{kt}(1-p)^{k-1} p \\
&= pe^t \sum_{k=1}^{\infty} (e^t(1-p))^{k-1} \\
&= \frac{pe^t}{1-e^t(1-p)}
\end{aligned}
$$

With the bound for t:

$$
e^t < \frac{1}{1-p} \Rightarrow t < -\ln(1-p)
$$

Hence, we have:

$$
\begin{aligned}
R(t) &= \ln\left(\frac{pe^t}{1-e^t(1-p)}\right) \\
R'(t) &= \frac{1-e^t(1-p)}{pe^t} \cdot \frac{pe^t(1-e^t(1-p)) + p(1-p)e^{2t}}{[1-e^t(1-p)]^2} \\
&= \frac{(1-e^t(1-p)) + (1-p)e^t}{1-e^t(1-p)} \\
&= 1 + \frac{(1-p)e^t}{1-e^t(1-p)} \\
R''(t) &= \frac{(1-p)e^t[1-e^t(1-p)] + [(1-p)e^t]^2}{[(1-e^t(1-p))]^2}
\end{aligned}
$$

By some calculation, we found $\mu = R'(0) = \frac{1}{p}$ and $\sigma^2 = \frac{(1-p)p+(1-p)^2}{p^2} = \frac{1-p}{p^2}$

(d)

Theorem 0.1. *The expected values of two random variables is multiplicative if* ***they are indenpendent****, i.e.,* $E[XY] = E[X][Y]$

Proof. If X and Y are independent (and $\mathcal{X}$ and $\mathcal{Y}$ are the sample spaces for X and Y respectively.), it holds that

$$p(x, y) = p(x)p(y) \quad \text{for all} \quad x \in \mathcal{X}, y \in \mathcal{Y}. \tag{1}$$

Applying this to the expected value for discrete random variables, we have

$$\begin{aligned} \mathrm{E}(XY) &= \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}}(x \cdot y) \cdot f_{X,Y}(x, y) \\ &= \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}}(x \cdot y) \cdot (f_X(x) \cdot f_Y(y)) \\ &= \sum_{x\in\mathcal{X}} x \cdot f_X(x) \sum_{y\in\mathcal{Y}} y \cdot f_Y(y) \\ &= \sum_{x\in\mathcal{X}} x \cdot f_X(x) \cdot \mathrm{E}(Y) \\ &= \mathrm{E}(X)\mathrm{E}(Y). \end{aligned}$$

And applying it to the expected value for continuous random variables, we have

$$\begin{aligned} \mathrm{E}(XY) &= \int_{\mathcal{X}}\int_{\mathcal{Y}}(x \cdot y) \cdot f_{X,Y}(x, y)\, dy\, dx \\ &= \int_{\mathcal{X}}\int_{\mathcal{Y}}(x \cdot y) \cdot (f_X(x) \cdot f_Y(y))\, dy\, dx \\ &= \int_{\mathcal{X}} x \cdot f_X(x) \int_{\mathcal{Y}} y \cdot f_Y(y)\, dy\, dx \\ &= \int_{\mathcal{X}} x \cdot f_X(x) \cdot \mathrm{E}(Y)\, dx \\ &= \mathrm{E}(X)\mathrm{E}(Y). \end{aligned}$$

□

Let Y be the r.v. for the number of trials to achiving exactly r successes.

We can express Y as the sum of r independent, identically distributed geometric random variables, p is the probability of getting success in a single test:

$$Y = X_1 + X_2 + \cdots + X_r$$

In this equation, X_1 is the number of trials to get the first success, X_2 is the number of additional trials to get the second success, and so on up to the r-th success.

As derived previously, the m.g.f for a single geometric r.v. is:

$$M_X(t) = \frac{pe^t}{1 - e^t(1-p)}$$

By Theorem 0.1, we can obtain the m.g.f for Y using following steps:

$$\begin{aligned} M_Y(t) &= E[e^{Yt}] \\ &= E[e^{(X_1+X_2+\cdots+X_r)}] \\ &= E[e^{X_1t} \cdot e^{X_2t} \cdot \cdots \cdot e^{X_rt}] \\ &= E[e^{X_1t}]E[e^{X_2t}] \ldots E[e^{X_rt}] \\ &= (\frac{pe^t}{1 - e^t(1-p)})^r \end{aligned}$$

Then let's derive $R(t)$, and its first and second derivatives:

$$\begin{aligned} R(t) &= \ln{(M_Y(t))} \\ &= r(\ln{(p)} + t - \ln{(1 - e^t(1-p))}) \\ R'(t) &= r(1 + \frac{(1-p)e^t}{1 - e^t(1-p)}) \\ R''(t) &= r\frac{(1-p)e^t[1 - e^t(1-p)] + [e^t(1-p)]^2}{[1 - e^t(1-p)]^2} \\ &= r\frac{(1-p)e^t}{[1 - e^t(1-p)]^2} \end{aligned}$$

By some easy calculation, we know $\mu = R'(0) = \frac{r}{p}$ and $\sigma^2 = R''(0) = \frac{r(1-p)}{p^2}$

2.6-11

3.1-8

3.2-4